

Introduction

This Appendix contains a summary of status and error messages, which status register bits are set when messages occur, and methods to avoid or eliminate most common SCPI errors.

Status and error messages

[Table B-1](#) summarizes status and error messages, which are stored in the Error Queue. Each message is preceded by a code number. Negative (-) numbers are used for SCPI defined messages, and positive (+) numbers are used for Keithley defined messages. Note that error and status conditions will also set specific bits in various status registers, as summarized in [Table B-1](#).

[Section 15](#) has detailed information on registers and queues. Briefly, you can use the following queries to obtain error and status information:

- :SYST:ERR? — reads Error Queue.
- *ESR? — reads Standard Event Status Register.
- :STAT:OPER? — reads Operation Event Register.
- :STAT:MEAS? — reads Measurement Event Register.
- :STAT:QUES? — reads Questionable Event Register.

NOTE SCPI confirmed messages are described in volume 2: *Command Reference of the Standard Commands for Programmable Instruments*. Refer to the :SYSTem:ERRor? command.

Table B-1
Status and error messages

Number	Error message	Event ¹	Status register ²	Bit
-440	Query UNTERMINATED after indefinite response	EE	Standard Event	2
-430	Query DEADLOCKED	EE	Standard Event	2
-420	Query UNTERMINATED	EE	Standard Event	2
-410	Query INTERRUPTED	EE	Standard Event	2
-363	Input buffer overrun	EE	Standard Event	3
-362	Framing error in program message	EE	Standard Event	3
-361	Parity error in program message	EE	Standard Event	3
-360	Communications error	EE	Standard Event	3
-350	Queue overflow	SYS	Standard Event	3
-330	Self-test failed	EE	Standard Event	3
-314	Save/recall memory lost	EE	Standard Event	3
-315	Configuration memory lost	EE	Standard Event	3
-285	Program syntax error	EE	Standard Event	4
-284	Program currently running	EE	Standard Event	4
-282	Illegal program name	EE	Standard Event	4
-281	Cannot create program	EE	Standard Event	4
-260	Expression error	EE	Standard Event	4
-241	Hardware missing	EE	Standard Event	4
-230	Data corrupt or stale	EE	Standard Event	4
-225	Out of memory	EE	Standard Event	4
-224	Illegal parameter value	EE	Standard Event	4
-223	Too much data	EE	Standard Event	4
-222	Parameter data out of range	EE	Standard Event	4
-221	Settings conflict	EE	Standard Event	4
-220	Parameter error	EE	Standard Event	4
-215	Arm deadlock	EE	Standard Event	4
-214	Trigger deadlock	EE	Standard Event	4
-213	Init ignored	EE	Standard Event	4
-212	Arm ignored	EE	Standard Event	4
-211	Trigger ignored	EE	Standard Event	4
-210	Trigger error	EE	Standard Event	4
-202	Settings lost due to rtl	EE	Standard Event	4
-201	Invalid while in local	EE	Standard Event	4
-200	Execution error	EE	Standard Event	4

Table B-1 (continued)
Status and error messages

Number	Error message	Event ¹	Status register ²	Bit
-178	Expression data not allowed	EE	Standard Event	5
-171	Invalid expression	EE	Standard Event	5
-170	Expression error	EE	Standard Event	5
-168	Block data not allowed	EE	Standard Event	5
-161	Invalid block data	EE	Standard Event	5
-160	Block data error	EE	Standard Event	5
-158	String data not allowed	EE	Standard Event	5
-154	String too long	EE	Standard Event	5
-151	Invalid string data	EE	Standard Event	5
-150	String data error	EE	Standard Event	5
-148	Character data not allowed	EE	Standard Event	5
-144	Character data too long	EE	Standard Event	5
-141	Invalid character data	EE	Standard Event	5
-140	Character data error	EE	Standard Event	5
-128	Numeric data not allowed	EE	Standard Event	5
-124	Too many digits	EE	Standard Event	5
-123	Exponent too large	EE	Standard Event	5
-121	Invalid character in number	EE	Standard Event	5
-120	Numeric data error	EE	Standard Event	5
-114	Header suffix out of range	EE	Standard Event	5
-113	Undefined header	EE	Standard Event	5
-112	Program mnemonic too long	EE	Standard Event	5
-111	Header separator error	EE	Standard Event	5
-110	Command header error	EE	Standard Event	5
-109	Missing parameter	EE	Standard Event	5
-108	Parameter not allowed	EE	Standard Event	5
-105	GET not allowed	EE	Standard Event	5
-104	Data type error	EE	Standard Event	5
-103	Invalid separator	EE	Standard Event	5
-102	Syntax error	EE	Standard Event	5
-101	Invalid character	EE	Standard Event	5
-100	Command error	EE	Standard Event	5
+000	No error	SE		

Table B-1 (continued)
Status and error messages

	Error message	Event ¹	Status register ²	Bit
	Measurement events:			
+100	Limit 1 failed	SE	Measurement Event	0
+101	Low limit 2 failed	SE	Measurement Event	1
+102	High limit 2 failed	SE	Measurement Event	2
+103	Low limit 3 failed	SE	Measurement Event	3
+104	High limit 3 failed	SE	Measurement Event	4
+105	Active limit tests passed	SE	Measurement Event	5
+106	Reading available	SE	Measurement Event	6
+107	Reading overflow	SE	Measurement Event	7
+108	Buffer available	SE	Measurement Event	8
+109	Buffer full	SE	Measurement Event	9
+110	Limit 4 failed	SE*	Measurement Event	10
+111	OUTPUT enable asserted	SE	Measurement Event	11
+112	Temperature limit exceeded	SE	Measurement Event	12
+113	Voltage limit exceeded	SE	Measurement Event	13
+114	Source in compliance	SE	Measurement Event	14
	Standard events:			
+200	Operation complete	SE	Standard Event	0
	Operation events:			
+300	Device calibrating	SE	Operation Event	0
+303	Device sweeping	SE	Operation Event	3
+305	Waiting in trigger layer	SE	Operation Event	5
+306	Waiting in arm layer	SE	Operation Event	6
+310	Entering idle layer	SE	Operation Event	10
	Questionable events:			
+408	Questionable Calibration	SE	Questionable Event	8
+414	Command Warning	SE	Questionable Event	14

* Contact check option only.

Table B-1 (continued)
Status and error messages

Number	Error message	Event ¹	Status register ²	Bit
	Calibration errors:			
+500	Date of calibration not set	EE	Standard Event	3
+501	Next date of calibration not set	EE	Standard Event	3
+502	Calibration data invalid	EE	Standard Event	3
+503	DAC calibration overflow	EE	Standard Event	3
+504	DAC calibration underflow	EE	Standard Event	3
+505	Source offset data invalid	EE	Standard Event	3
+506	Source gain data invalid	EE	Standard Event	3
+507	Measurement offset data invalid	EE	Standard Event	3
+508	Measurement gain data invalid	EE	Standard Event	3
+509	Not permitted with cal locked	EE	Standard Event	3
+510	Not permitted with cal un-locked	EE	Standard Event	3
	Lost data errors:			
+601	Reading buffer data lost	EE	Standard Event	3
+602	GPIO address lost	EE	Standard Event	3
+603	Power-on state lost	EE	Standard Event	3
+604	DC calibration data lost	EE	Standard Event	3
+605	Calibration dates lost	EE	Standard Event	3
+606	GPIO communication language lost	EE	Standard Event	3
	Communication errors:			
+700	Invalid system communication	EE	Standard Event	3
+701	ASCII only with RS-232	EE	Standard Event	3
	Additional command execution errors:			
	Illegal with storage active			
+800	Insufficient vector data	EE	Standard Event	4
+801	OUTPUT blocked by output enable	EE	Standard Event	4
+802	Not permitted with OUTPUT off	EE	Standard Event	4
+803	Expression list full	EE	Standard Event	4
+804	Undefined expression exists	EE	Standard Event	4
+805	Expression not found	EE	Standard Event	4
+806	Definition not allowed	EE	Standard Event	4
+807	Expression cannot be deleted	EE	Standard Event	4
+808	Source memory location revised	EE	Standard Event	4
+809	OUTPUT blocked by Over Temp	EE	Standard Event	4
+810		EE	Standard Event	4

Table B-1 (continued)
Status and error messages

Number	Error message	Event ¹	Status register ²	Bit
+811	Not an operator or number	EE	Standard Event	4
+812	Mismatched parenthesis	EE	Standard Event	4
+813	Not a number of data handle	EE	Standard Event	4
+814	Mismatched brackets	EE	Standard Event	4
+815	Too many parenthesis	EE	Standard Event	4
+816	Entire expression not parsed	EE	Standard Event	4
+817	Unknown token	EE	Standard Event	4
+818	Error parsing mantissa	EE	Standard Event	4
+819	Error parsing exponent	EE	Standard Event	4
+820	Error parsing value	EE	Standard Event	4
+821	Invalid data handle index	EE	Standard Event	4
+822	Too small for sense range	EE	Standard Event	4
+823	Invalid with source read-back on	EE	Standard Event	4
+824	Cannot exceed compliance range	EE	Standard Event	4
+825	Invalid with auto-ohms on	EE	Standard Event	4
+826	Attempt to exceed power limit	EE	Standard Event	4
+827	Invalid with ohms guard on	EE	Standard Event	4
+828	Invalid on 1 amp range	EE	Standard Event	4
+829	Invalid on 1kV range	EE	Standard Event	4
+830	Invalid with INF ARM:COUNT	EE	Standard Event	4
+831	Invalid in Pulse Mode	EE	Standard Event	4
+900	Internal System Error	EE	Standard Event	3

1. EE = Error Event
SE = Status Event
SYS = System Error Event
2. Use following queries to read status registers:
Standard Event: *ESR?
Operation Event: STAT:OPER?
Measurement Event: STAT:MEAS?
Questionable Event: STAT:QUES?

Eliminating common SCPI errors

There are three SCPI errors that occur more often than any others:

- -113, "Undefined header"
- -410, "Query INTERRUPTED"
- -420, "Query UNTERMINATED"

The following paragraphs discuss the most likely causes for these errors and methods for avoiding them.

-113, "Undefined header"

This error indicates that the command you sent to the instrument did not contain a recognizable command name. The most likely causes for this error are:

- *Missing space between the command and its parameter.* There must be one or more spaces (blanks) between the command and its parameter. For example:

:SENS:VOLT:DC:RANG100 Incorrect (no space between command and parameter)

:SENS:VOLT:DC:RANG 100Correct

- *Improper short or long form.* Check the command list in Section 18 of this manual.
- *Blanks (spaces) within the command name.* For example:

:SYST :ERR? Incorrect (space between :SYST and :ERR?)

:SYST:ERR? Correct

-410, "Query INTERRUPTED"

This error occurs when you have sent a valid query to the instrument, and then send it another command or query, or a Group Execute Trigger (GET) before it has had a chance to send the entire response message (including the line-feed/EOI terminator). The most likely causes are:

- *Sending a query to the instrument and then sending another command or query before reading the response to the first query.* For example, the following sequence of commands will cause an error -410:

:SYST:ERR?

*OPC?

This sequence generates an error because you must read the response to :SYST:ERR? before sending the *OPC? query.

- *Incorrectly configured IEEE-488 driver.* The driver must be configured so that when talking on the bus it sends line-feed with EOI as the terminator, and when listening on the bus it expects line-feed with EOI as the terminator. See the reference manual for your particular IEEE-488 interface.

-420, "Query UNTERMINATED"

This error occurs when you address the instrument to talk, and there is no response message to send. The most likely causes are:

- *Not sending a query.* You must send a valid query to the instrument before addressing it to talk.
- *Sending an invalid query.* If you have sent a query and still get this error, make sure that the instrument is processing the query without error. For example, sending an ill-formed query that generates an error -113, "Undefined header" and then addressing the instrument to talk will generate an error -420, "Query UNTERMINATED" as well.
- *Valid query following an invalid command.* This situation can occur when you send multiple commands or queries (program message units) within one command string (program message). When the SourceMeter detects an error in a program message unit, it discards all further program message units until the end of the string; for example:

```
:SENS:DATE?; :SENS:FUNC?
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In the above program message, the program message unit :SENS:DATE? will generate error -113, "Undefined header," and the SourceMeter will discard the second program message unit :SENS:FUNC? even though it is a valid query.